



A multi-gene blood-based methylation assay for early diagnosis of colorectal cancer

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Background: Early detection for colorectal cancer (CRC) can enhance the patient prognosis. We aimed to validate the combined multi-gene detection in plasma of *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* for early diagnosing of CRC in this prospective study.

Methods: Overall, 124 participants including 45 CRC patients, 8 advanced adenoma patients, 34 small polyp patients, and 37 normal controls who underwent colonoscopy were enrolled. The carcinoembryonic antigen (CEA) test and methylation tests for *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* were performed. Sensitivity, specificity, and the area under the curve (AUC) of the receiver operating characteristic (ROC) curve were utilized to evaluate the diagnostic value of each biomarker. Additionally, the association between the positive rates of methylated *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* and the clinicopathological characteristics of CRC was also analyzed.

Results: The positive detection rate of multi-gene methylation in CRC patients was 86.67%, for stage I and stage II patients, the positive rates were 90.91% and 87.50%, both of which were significantly higher than CEA, which had rates of 55.56%, 18.18% and 56.25% for the corresponding stages. In patients with advanced adenomas and small polyps, the positive rates for the four-gene combined test were 62.50% and 52.94%, respectively, which were markedly higher than the CEA rates of 12.50% and 14.71%. AUC of the ROC curve indicated that the diagnostic value of the multi-gene test for CRC was superior to that of any single gene. Correlation analysis revealed that the positive rate of the test was not affected by patients' clinicopathological characteristics.

Conclusions: A combination of methylated *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* test has the potential for early diagnosis of CRC patients, advanced adenoma patients, and small polyp patients.

Keywords: Colorectal cancer (CRC); early diagnosis; DNA methylation; multi-gene; blood-based

Submitted Apr 30, 2024. Accepted for publication Oct 31, 2024. Published online Dec 20, 2024.

doi: 10.21037/tcr-24-729

View this article at: <https://dx.doi.org/10.21037/tcr-24-729>

Introduction

Colorectal cancer (CRC), also known as bowel cancer, is one of the malignant tumors that pose a significant threat to human health worldwide. Its incidence and mortality

rates remain persistently high, contributing substantially to the increasing global cancer burden. According to relevant research, CRC currently ranks third in incidence among all cancers globally and second in mortality (1). Statistics

from China's cancer data in 2020 indicate that CRC holds the second position in cancer incidence and the fifth in mortality within the country (2). However, due to the often atypical and subtle early symptoms of CRC, patients are typically diagnosed only when overt symptoms, such as rectal bleeding, changes in bowel habits, and alterations in stool characteristics, manifest. By this time, most cases have already progressed to advanced stages, missing the optimal treatment window, which results in poor prognosis and reduced 5-year survival rates (3). Therefore, identifying a screening method with high sensitivity and specificity for early CRC detection is of paramount importance.

Various techniques, including the fecal occult blood test (FOBT), serum carcinoembryonic antigen (CEA) test, colonoscopy, and computed tomography (CT) simulation, have been used in clinics. FOBT and colonoscopy are the primary detection methods, however, they are hindered by

low specificity and sensitivity, as well as high invasiveness, making it difficult to widely promote CRC screening. Furthermore, FOBT can be easily affected by diet, which can compromise the accuracy of test outcomes. While colonoscopy is effective, it is time-consuming, labor-intensive, and may cause discomfort to patients, as well as being expensive, making its large-scale promotion challenging (4). In the current healthcare system, the responsibility for CRC screening primarily rests with medical examination centers. Unfortunately, there are significant deficiencies in this critical aspect of CRC prevention. Most examination centers tend to rely on initial screenings using tumor markers such as CEA, while the use of more comprehensive tests like FOBT and colonoscopy, remains relatively low. Additionally, there are considerable variations in the technical proficiency of tumor marker detection across different centers, which undoubtedly increases the risk of missed diagnoses among CRC patients and hinders the timely detection of numerous potential cases. Therefore, there is an urgent need for a novel screening method that is efficient, cost-effective, and easily accessible to the public.

Non-invasive detection methods utilizing molecular markers have increasingly emerged as powerful tools for the early diagnosis of CRC. In particular, DNA methylation, a common and stable epigenetic modification, shows abnormalities not only in the early stages of CRC but can also be detected in bodily fluids. This characteristic makes it a significant molecular marker for the early diagnosis of CRC and has been confirmed to play a role in the initial development of various tumors (5). Numerous genes have been identified as effective indicators for diagnosing early CRC (6,7), including those studied here, such as *Septin9*, *SDC2*, *IKZF1*, and *KCNQ5*, as well as other genes like *SFRP1* (8) and *C9orf50* (9). Most diagnoses rely on single or double gene markers, while the use of triple or even quadruple gene combinations remains rare. Single gene detection often has a low positive rate, which limits its effectiveness in early screening of CRC. For example, while *Septin9* methylation has a high detection rate in CRC, it has a low detection rate in polyps and adenomas, at only 17.1% (10). In fact, some researchers have found the diagnostic value of plasma *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* for CRC, but no research has focused on the potential of a combined four-gene methylation test for CRC early detection. The aim of this study was to investigate and explore the significance of multi-gene methylation detection in plasma circulating tumor DNA (ctDNA) for

Highlight box

Key findings

- Our prospective study indicates that a multi-gene methylation assay for *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* in plasma significantly surpasses the detection rate of the conventional carcinoembryonic antigen (CEA) test in identifying colorectal cancer (CRC), demonstrating a markedly higher sensitivity rate of 86.67% compared to CEA's 55.56%. Remarkably, for early-stage patients (stages I and II), our assay achieves striking positivity rates of 90.91% and 87.50%, respectively.

What is known and what is new?

- CRC is well-known for its potential to be prevented through early detection. However, traditional methods such as CEA testing and colonoscopies have limitations in sensitivity and accessibility. Single-gene methylation tests have shown promise but often fall short in providing the sensitivity required for the detection of early-stage cancer.
- This study introduces a novel four-gene panel that not only improves detection rates but also provides a non-invasive and highly sensitive alternative for early-stage diagnosis of CRC.

What is the implication, and what should change now?

- The diagnostic precision of our multi-gene methylation assay has significant implications for the early detection of CRC. This advancement could not only increase the chances of early detection but also alleviate the burden of invasive procedures, providing patients with a more receptive and effective approach to cancer screening. Therefore, we must diligently pursue the identification of additional genomic markers that could be pivotal for diagnostic purposes. Our goal is to fully explore the potential of epigenetic research in the field of CRC.

Table 1 Primer and probe sequences

Genes	Forward (5' to 3')	Reverse (5' to 3')	Probe sequence
<i>Septin9</i>	ATAATCCCATCCAA	ATTGTTGTTTATTAGTTAT	JOE-TTAACCGCGAAATCCGAC-BHQ
<i>SDC2</i>	ATTAATAAGTGAGAGGGCGT	AAACTCGAAAACCTCGAA	FAM-GCGTAGGAGGAAGCGA-BHQ
<i>KCNQ5</i>	TTTGTTGGGGAAGTC	TTAACGTTACGCCGA	Cy5-CGAGTAGTTAGAGTTG-BHQ
<i>IKZF1</i>	CGTATTTTTCGTGTTTC	CACCTCTCGACCG	JOE-ATCGGAGTAGCGATTCTGGGAG-BHQ
<i>ACTB</i>	GGAGGAGGTTTAGTAAGTT	AAAACCTACTCCTCCCTTAA	ROX-AACACACAATAACAAACACA-BHQ

the early diagnosis of CRC, with the goal of achieving a low-cost, convenient, and more accurate screening method that is urgently needed in China, thereby promoting early CRC screening in the country. We present this article in accordance with the STARD reporting checklist (available at <https://tcr.amegroups.com/article/view/10.21037/tcr-24-729/rc>).

Methods

Study design

This is a prospective study that enrolled 124 participants from January 2023 to August 2024 in the Department of Anorectology of the Second Hospital of Tianjin Medical University.

Participants

The participants were categorized into four groups: the CRC group, the advanced adenoma group, the small polyps, and the control group. All blood samples used in this study were collected at the Second Hospital of Tianjin Medical University (Tianjin, China).

The inclusion criteria for this study are those who underwent colonoscopy and provided complete clinical information. Additionally, the pathological results of the samples were examined and confirmed by at least two attending pathologists. The diagnostic criteria referred to the diagnostic criteria of colorectal polyps in the Clinical Guidelines for Gastrointestinal Diseases. Those with a history of radiotherapy for CRC or any other malignant tumors were excluded from the study. This research was approved by the Ethics Committee of the Second Hospital of Tianjin Medical University (No. KY2023K207), and all participants signed informed consent forms. The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013).

Test methods

DNA isolation and methylation testing

K₂EDTA anticoagulation blood collection tube (10 mL specification single-use free DNA preservation tube, Kangwei Century, Taizhou, China, SUMCI 20192220059) was used to collect venous blood sample.

Free DNA in the plasma sample was extracted using the VAHTS Serum/Plasma Circulating DNA Kit (N902-00, Lot#30825) from Novozymes, with a 2 mL sample volume. The final elution volume was 100 μ L. Then, 60 μ L of the extracted product was treated with bisulfite using Epigenetics' EZ DNA Methylation-Gold Kit, D5006, Sulfurization Kit, and 35 μ L was used for the final elution. The operation procedure was described in the instruction manual.

Quantitative polymerase chain reaction (PCR) reactions were prepared according to the sample volume. The reaction conditions were: pre-denaturation at 95 $^{\circ}$ C for 5 minutes, denaturation at 95 $^{\circ}$ C for 15 seconds, annealing and extension at 57 $^{\circ}$ C for 60 seconds, 40 cycles, and finally, fluorescence values were recorded. Primer and probe sequences are shown in *Table 1*.

After the reaction was completed, the threshold and baseline of each channel were adjusted according to the situation. The Ct value of each signal channel in the reaction well of each sample was determined, and the difference between the Ct value of each target gene and the Ct value of the *ACTB* gene (internal reference control) was calculated as Δ Ct [Δ Ct = Ct (target gene) – Ct (*ACTB*)]. If the Ct value of *ACTB* was greater than 34, the result was considered invalid. If no signal was detected from PCR, the Ct value was assigned as 40 (the maximum number of PCR cycles). The methylation of *Septin9*, *SDC2*, *KCNQ5* and *IKZF1* were determined by calculating the Δ Ct compared to *ACTB*. when Δ Ct $\leq$ 7.49, Δ Ct $\leq$ 7.445, Δ Ct $\leq$ 5.695, and Δ Ct $\leq$ 5.96, the genes *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* in the test sample were identified as positively methylated.

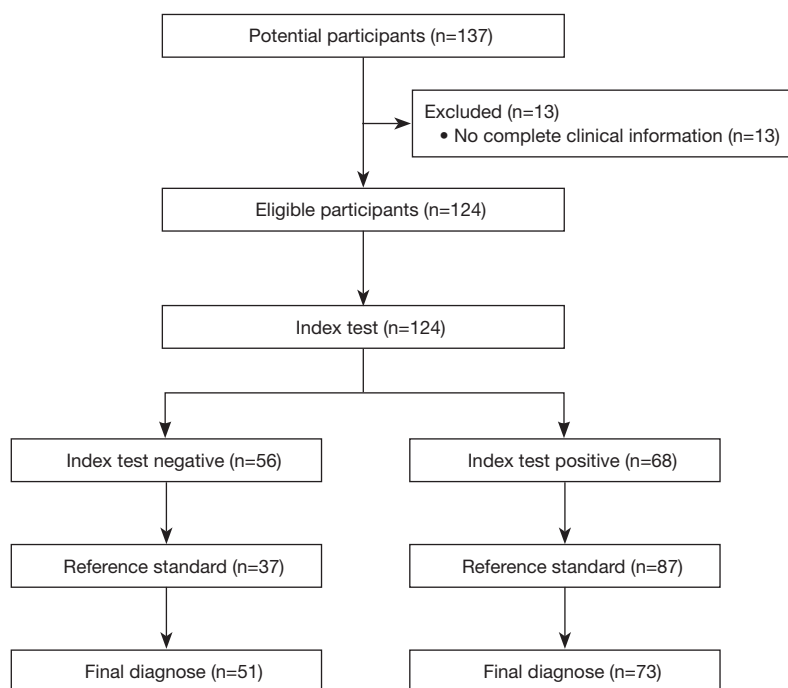


Figure 1 Flow of participants through the study.

Positive methylation of any gene was defined as combined positive of four genes.

Methylation detection in HCT116 and 293T cell lines

The HCT116 cell line, which exhibits methylation in all four genes—*Septin9*, *SDC2*, *KCNQ5*, and *IKZF1*, was used as a positive control. In contrast, the 293T cell line, known for having no methylation in any of these four genes, served as a negative control. Quantitative PCR was performed to assess the methylation status of the *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* genes.

CEA testing

CEA levels in serum were measured by Roche Cobas 8000 electrochemiluminescence instrument (Shanghai, China). A CEA level of ≤ 5 ng/mL was the normal reference range for patients with a history of smoking, while a CEA level of ≤ 3.5 ng/mL was the normal reference range for patients without a smoking history. Samples with values exceeding the upper limit of the normal range were classified as positive.

Statistical analyses

SPSS software (version 26.0; IBM Corp., Armonk, NY,

USA) was applied for statistical analysis. Categorical data were analyzed using the Chi-squared test. Comparisons among more than two groups were performed using a one-way analysis of variance (ANOVA) test. Sensitivity, specificity, and the area under the curve (AUC) were used to evaluate the performance of methylated *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* in predicting CRC. $P < 0.05$ was considered to be statistically significant.

Results

Participants

The flow diagram for recruitment is illustrated in *Figure 1*. The 124 participants were divided into four groups. The CRC group comprised 45 CRC patients, including 31 males and 14 females, with ages ranging from 41 to 85 years old and an average age of 69.09 ± 9.14 years. The advanced adenoma group included 8 patients, comprising 5 males and 3 females, aged between 34 and 69 years, with an average age of 59.88 ± 13.79 years. The small polyps group contained 34 patients, with an equal distribution of 17 males and 17 females, aged from 43 to 85 years, and an average age of 60.35 ± 9.83 years. The control group consisted of 37 cases, including 23 males and 14 females, aged from

Table 2 Clinical and pathological data of patients and healthy controls

Characteristics	All, n	Male, n (%)	Female, n (%)	Age (years), mean ± SD
CRCs	45	31 (68.89)	14 (31.11)	69.09±9.14
Pathologic stage (TNM)				
Stage I	11	7 (63.64)	4 (36.36)	68.27±7.02
Stage II	16	10 (62.50)	6 (37.50)	68.25±11.74
Stage III	12	9 (75.00)	3 (25.00)	69.58±9.36
Stage IV	6	5 (83.33)	1 (16.67)	71.83±4.26
Advanced adenoma	8	5 (62.50)	3 (37.50)	59.88±13.79
Small polyp	34	17 (50.00)	17 (50.00)	60.35±9.83
Healthy controls	37	23 (62.16)	14 (37.84)	43.65±13.16

CRC, colorectal cancer; TNM, tumor-node-metastasis; SD, standard deviation.

Table 3 The Ct value of *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* in HCT116 and 293T cell lines

Cell line	<i>Septin9</i>	<i>SDC2</i>	<i>ACTB</i>	<i>KCNQ5</i>	<i>IKZF1</i>
HCT116	23.60	26.78	25.76	25.00	24.73
293T	NoCt	NoCt	26.46	NoCt	NoCt
Control	NoCt	NoCt	NoCt	NoCt	NoCt

15 to 72 years and an average age of 43.65±13.16 years, as shown in *Table 2*.

Methylation detection of Septin9, SDC2, KCNQ5, and IKZF1 in HCT116 and 293T cell lines

As shown in *Table 3*, significant differences in the Ct value of *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* were observed between HCT116 and 293T cell lines, demonstrating the validity of the method used in this study.

Positive detection rates of methylated Septin9, SDC2, KCNQ5, IKZF1 and CEA

When methylated *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* were combined for CRC detection, the positivity rate was higher than that of the healthy group. In CRC patients, the combined methylation test of these four genes demonstrated a positivity rate of 86.67%, surpassing that of CEA, which was 55.56%. Notably, in patients with stage I and stage II CRC, the positivity rates of the four-gene combination (blue bar) were 90.91% and

87.50%, compared to CEA's rates of 18.18% and 56.25% (green bar). When these five indicators were tested in combination, the overall positivity rate for CRC reached 93.33%. In advanced adenoma and small polyp patients, the positive rates for the four-gene combined test were 62.50% and 52.94% (blue bar), respectively, which were markedly higher than those of CEA (12.50% and 14.71% respectively) (green bar) (*Figure 2*, *Table 4*). The combination was considered positive if any one of the aforementioned genes showed positivity.

Diagnostic value of methylated Septin9, SDC2, KCNQ5, and IKZF1 for CRC

Receiver operating characteristic (ROC) analyses indicated that when *Septin9*, *SDC2*, *KCNQ5*, and *IKZF1* were detected individually, the AUC (sensitivity and specificity) values were 0.787 (71.10%, 75.90%), 0.615 (48.90%, 75.90%), 0.730 (66.70%, 78.50%), and 0.670 (46.70%, 88.60%), respectively. While the AUC (sensitivity and specificity) was 0.809 (66.70%, 83.50%) in the four-gene combination test, as shown in *Table 5* and *Figure 3*.

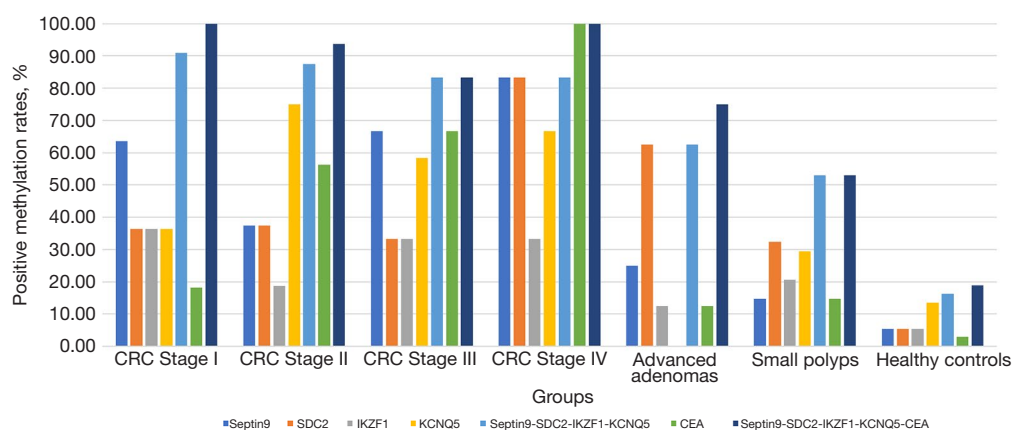


Figure 2 Positive methylation rates in the groups. CRC, colorectal cancer; CEA, carcinoembryonic antigen.

Table 4 Comparison of positive methylation rates in different groups

Characteristics	N	Septin9	SDC2	IKZF1	KCNQ5	CEA	Septin9-SDC2-IKZF1-KCNQ5	Septin9-SDC2-IKZF1-KCNQ5-CEA
CRCs	45	26 (57.78)	19 (42.22)	13 (28.89)	27 (60.00)	25 (55.56)	39 (86.67)	42 (93.33)
Stage I	11	7 (63.64)	4 (36.36)	4 (36.36)	4 (36.36)	2 (18.18)	10 (90.91)	11 (100.00)
Stage II	16	6 (37.50)	6 (37.50)	3 (18.75)	12 (75.00)	9 (56.25)	14 (87.50)	15 (93.75)
Stage III	12	8 (66.67)	4 (33.33)	4 (33.33)	7 (58.33)	8 (66.67)	10 (83.33)	10 (83.33)
Stage IV	6	5 (83.33)	5 (83.33)	2 (33.33)	4 (66.67)	6 (100.00)	5 (83.33)	6 (100.00)
Advanced adenomas	8	2 (25.00)	5 (62.50)	1 (12.50)	0 (0.00)	1 (12.50)	5 (62.50)	6 (75.00)
Small polyps	34	5 (14.71)	11 (32.35)	7 (20.59)	10 (29.41)	5 (14.71)	18 (52.94)	18 (52.94)
Healthy controls	37	2 (5.41)	2 (5.41)	2 (5.4)	5 (13.51)	1 (3.00)	6 (16.22)	7 (18.92)
P value		<0.001	<0.001	0.04	<0.001	<0.001	<0.001	<0.001

Data are presented as number or n (%). CRC, colorectal cancer; CEA, carcinoembryonic antigen.

Analysis of the correlation between a four-gene combination test and clinicopathologic characteristics of CRC patients

No statistically significant correlation was found between the positivity rate of the four-gene combination test and the age, gender, history of diabetes, primary tumor location, gross type, tumor-node-metastasis (TNM) stage, or degree of differentiation of CRC patients ($P>0.05$), as shown in Table 6.

Discussion

CRC primarily presents as sporadic cases rather than showing familial genetic clustering. The progression of this disease is notably protracted, often involving a transition from benign adenomas to cancer. This process

typically spans a latency period of 5 to 10 years, during which subtle cellular alterations gradually develop into detectable tumors. Studies have shown that individuals with high-grade adenomas have a significantly increased risk of developing CRC compared to those without adenomas, and the mortality rate associated with CRC is also markedly elevated (11,12). Given distinct features of disease progression, CRC is one of the few malignant tumors that can be screened to reduce morbidity and mortality rates.

The gold standard for CRC screening is the invasive method colonoscopy. It requires patients to prepare their intestines in advance, which can lead to poor compliance. In addition, there are risks of bleeding, perforation and infection. Under current medical conditions colonoscopy equipment and specialized medical personnel cannot meet

Table 5 Analysis of the diagnostic value of each single and combined index for colorectal cancer

Genes	Sensitivity (%)	Specificity (%)	AUC (%)	95% CI of AUC	P value
<i>Septin9</i>	71.10	75.90	0.787	0.703–0.871	<0.001
<i>SDC2</i>	48.90	75.90	0.615	0.507–0.723	0.03
<i>IKZF1</i>	46.70	88.60	0.670	0.565–0.774	0.002
<i>KCNQ5</i>	66.70	78.50	0.730	0.636–0.823	<0.001
<i>Septin9-SDC2-IKZF1-KCNQ5</i>	66.70	83.50	0.809	0.730–0.888	<0.001

AUC, area under the curve; CI, confidence interval.

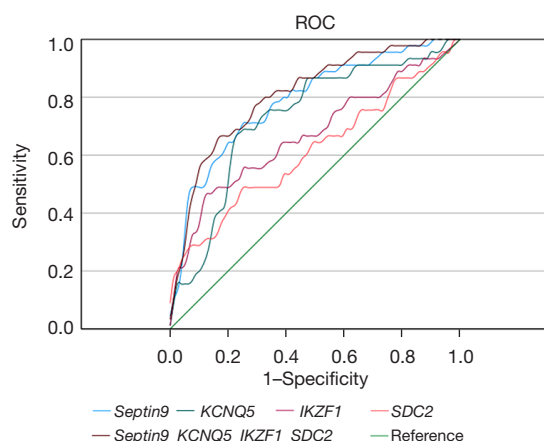


Figure 3 Diagnostic performance of *Septin9*, *SDC2*, *KCNQ5*, *IKZF1*, and the four-gene combination. The ROC results for the diagnostic performance of methylated *Septin9* (blue line), methylated *SDC2* (pink line), methylated *KCNQ5* (green line), methylated *IKZF1* (purple line), and the four-gene combination methylated (brown line) in patients with CRC. The AUC values were 0.787 (95% CI: 0.703–0.871) for *Septin9*, 0.615 (95% CI: 0.507–0.723) for *SDC2*, 0.730 (95% CI: 0.636–0.823) for *KCNQ5*, 0.670 (95% CI: 0.565–0.774) for *IKZF1*, and 0.809 (95% CI: 0.730–0.888) for the four-gene combination. ROC, receiver operating characteristic; CRC, colorectal cancer; AUC, area under the curve; CI, confidence interval.

the demand for large-scale screening of CRC. Unlikely colonoscopy, fecal testing is convenient for sampling, but the sensitivity in fecal immunochemical test (FIT) or FOBT in adenomas is low, which limits its application in CRC screening (13,14). Besides, a survey on sample preference during CRC screening found that about 80% of participants preferred to provide blood samples (15). Blood testing is convenient, easy to implement, and well-complied, which helps to increase the participation of population screening.

DNA methylation occurs in CpG dinucleotides within

higher eukaryotic genomes and is an extensive studied epigenetic phenomenon. Research has demonstrated that DNA methylation is an early event in the development of various tumors, and the methylation of specific genes can serve as molecular markers for early tumor screening (16). Some environmental factors can alter the hyper/hypomethylation of cancer suppressor gene promoters, proto-oncogene promoters, and the whole genome, causing low/high expression or gene mutation of related genes, thereby exerting oncogenic or anticancer effects. Jiang *et al.* have found that air pollution exposure is associated with the incidence of CRC. Air pollution exposure may affect the occurrence and development of CRC through epigenetic regulation of TMBIM1/PNKD, CXCR5 and TMEM110 (17).

A variety of single-gene methylation detection have been reported, but most of them tend to have limitations in the early detection of adenomas or CRC. Multi-gene methylation detection has emerged a future trend. Relevant studies have reported on the application of different combined gene methylation assays. For example, the sensitivity of the *SFRP2-TFPI2-NDRG4-BMP3* combined detection for CRC was 94.3%, which was about 30% higher than that of the single target (18); Ahlquist *et al.* found that compared with *Septin9* (60% sensitivity) for early diagnosis of CRC, the sensitivity of 5-gene *BMP3-NDRG4-TFPI2-KRAS-β-actin* gene combined detection could reach 87% (19). Therefore, the combination of genes is crucial for CRC diagnosis.

Septin9 is a widely used molecular marker for CRC screening. However, studies have indicated that its detection rate for early-stage CRC and advanced adenomas remains inadequate when used isolation (10,20). The expression level of *SDC2* in human colorectal adenocarcinoma tissue was higher than that in normal epithelial tissue (21). Abnormal methylation of *SDC2* at CpG sites is commonly observed in tumor tissue from the majority of CRC patients. The sensitivity and specificity of *SDC2* methylation in fecal

Table 6 Association of plasmatic methylated genes positivity with clinicopathological characteristics of colorectal cancer patients

Characteristic	No. of patients	Septin9-SDC2-IKZF1-KCNQ5		
		Sensitivity (%)	χ^2	P value
Age			0.067	0.80
<60 years	6	83.33		
≥60 years	39	87.18		
Gender			3.127	0.07
Male	31	80.65		
Female	14	100.00		
History of diabetes			0.124	0.73
Yes	10	90.00		
No	35	85.71		
Tumor location			0.016	0.90
Colon	14	85.71		
Rectum	31	87.10		
Gross type			0.495	0.48
Ulcerative	42	85.71		
Other	3	100.00		
Pathologic stage (TNM)			0.288	0.59
I–II	27	88.89		
III–IV	18	83.33		
Tumor differentiation			1.275	0.26
Low degree	7	100.00		
Moderate degree	38	84.21		

TNM, tumor-node-metastasis.

DNA for diagnosing CRC range from 77.4% to 81.1% and from 88.2% to 98%, respectively (22,23). It has been reported that the sensitivity of *SDC2* gene in blood DNA detection of CRC patients was 87.0% and the specificity was 95.2% (24). *IKZF1* has been implicated in tumor growth and invasion, and its methylation has been detected in a variety of tumors (25). It has been reported that the blood *BCAT1/IKZF1* double-gene methylation test is more sensitive and specific than the FIT test in the diagnosis of CRC (26). Jensen *et al.* found that methylated *KCNQ5* exhibited good performance in the early CRC detection in blood samples, with a sensitivity of 87.5% (27). The four genes showed specific advantages in the detection of CRC patients, however, the combined diagnostic performance of these four genes has not been reported.

In this research, we found that when *Septin9*, *SDC2*, *IKZF1*, and *KCNQ5* were tested in combination, the positivity rate of CRC, particularly in patients with stage I and stage II, advanced adenomas and small polyps, was significantly higher than that of the tumor marker CEA test. The diagnostic value of the combined four-gene test for CRC was found to be superior to that of any single-gene test by analyzing the ROC curve. Additionally, we found no correlation between the methylation status of the *Septin9*, *SDC2*, *IKZF1*, and *KCNQ5* genes in plasma ctDNA and factors such as gender, age, history of diabetes, primary tumor location, gross type, TNM stage, and degree of differentiation in CRC patients ($P>0.05$). This indicates that the markers of the four gene methylations exhibit strong stability in CRC patients, making them equally

effective for screening across different ages and pathological stages. Furthermore, multi-gene methylation detection can facilitate the early diagnosis of CRC and improve patient compliance. If the result is positive, an endoscopic examination can be conducted to confirm the diagnosis, and if it is negative, regular follow-up observations can be implemented.

Conclusions

The multi-gene detection of *Septin9*, *SDC2*, *IKZF1*, and *KCNQ5* demonstrates a high diagnostic positivity rate in patients with CRC, as well as in those with advanced adenomas and small polyps. This makes it a promising indicator for CRC and precancerous lesion screening in clinical practice. However, it has a few limitations, for example, the number of patients with advanced adenomas patients is small; the mean age of the healthy group is younger than that of CRC patients and the false positivity rate of the four-gene methylation test in the healthy group is higher than that of CEA.

While multi-gene methylation detection can overcome the limitations of poor sensitivity of single-gene detection, there are several challenges in its future applications: (I) further verify the most suitable target gene; (II) community and health checkups based on multi-gene methylation testing have not been performed; (III) combine with other molecular detection methods to address the limitations of polygenic methylation detection. To draw more convincing conclusions, it is necessary to conduct a multi-center comparative study. Only by doing so can we effectively demonstrate the clinical significance of the multi-gene detection of *Septin9*, *SDC2*, *IKZF1*, and *KCNQ5* in CRC screening.

Acknowledgments

Funding: This study was supported by grants from the Tianjin Natural Science Foundation (No. 21JCYBJC01730) and the Youth Foundation of the Second Hospital of Tianjin Medical University (Nos. 2019ydey02, 2020ydey20), which paid all editing and publication fees.

Footnote

Reporting Checklist: The authors have completed the STARD reporting checklist. Available at <https://tcr.amegroups.com/article/view/10.21037/tcr-24-729/rc>

Data Sharing Statement: Available at <https://tcr.amegroups.com/article/view/10.21037/tcr-24-729/dss>

Peer Review File: Available at <https://tcr.amegroups.com/article/view/10.21037/tcr-24-729/prf>

Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://tcr.amegroups.com/article/view/10.21037/tcr-24-729/coif>). Y.X. and X.L. are employees of Molecular Diagnostic Engineering Technology Research Center of Zhengzhou. The other authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013). This research was approved by the Ethics Committee of the Second Hospital of Tianjin Medical University (No. KY2023K207), and all participants signed informed consent forms.

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Cite this article as: Xu Y, Qu H, Liang R, Li M, Li M, Li X, Wang Z. A multi-gene blood-based methylation assay for early diagnosis of colorectal cancer. *Transl Cancer Res* 2024;13(12):6699-6708. doi: 10.21037/tcr-24-729